DataBase Assignment

EE 470 - Introduction to the Internet of Things- Fall / Version: V2

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This project must be done individually!

Link to this document:
https://docs.google.com/document/d/1zhLaeiWjtqfpCTeJzd_n543yl6_SxahBxWd4lHh4srY/edit?usp=sharing

Please make sure you submit your assignment individually!

DataBase Assignment

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Assignment Overview

The purpose of this exercise is to become familiar with MySQL database and its commands.

1. Learning Objectives

- Set Up an online web page
- Learn about SQL
- Practice SQL commands
- Create a Table for sensor data
- Using Github.

2. Required Materials & Reference Information

- Signup for https://hostinger.com
- database tutorial available on the course GitHub
- Tutorial on creating an account with hostinger.com
- You can find the longitude and latitude format here
- Good tutorial: https://www.tutorialspoint.com/mariadb/mariadb_introduction.htm

3. Experiment

This experiment has TWO parts.

Part 1:
In this part of the experiment you need to signup for an account at https://hostinger.com
This will allow you to have your personal web page and database.

Part 2:
In the second part of the experiment you will need to access the database using a PHP file and visualize the data using the Chartjs library.

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PART 1

1- A Short Introduction to Databases

A database is a systematic or organized collection of related information that is stored in such a way that it can be easily accessed, retrieved, managed, and updated. It is where all data is stored, very much like a library that houses a wide range of books from different genres. Think of data as books.

In a database, you can organize the data in rows and columns in the form of a table. Indexing the data makes it easy to find and retrieve it again as and when required. Many websites on the World Wide Web are managed with the help of databases. To create a database so that the data is accessible to users through only one set of software programs, database handlers are used.

MySQL, SQL Server, MongoDB, Oracle Database, PostgreSQL, Informix, Sybase, etc. are all examples of different databases. These modern databases are managed by DBMS. Structured Query Language, or SQL as it is more widely known, is used to operate on the data in a database.

MariaDB is a popular fork of MySQL created by MySQL's original developers. It grew out of concerns related to MySQL's acquisition by Oracle. It offers support for both small data processing tasks and enterprise needs. It aims to be a drop-in replacement for MySQL requiring only a simple uninstall of MySQL and an install of MariaDB. MariaDB offers the same features of MySQL and much more.

For more information about Databases read this: https://intellipaat.com/blog/what-is-database/

1.1. Create an Account

A- Carefully review the Tutorial on creating an account with hostinger.com. Create a very simple webpage, as described in the tutorial. Make sure your web page is working. Add your name to your webpage. We will call this web page Internet of Things (IoT). We will use this web page to place links to all your future assignments.

https://antoniokassis.com/firstpage.html
Link to your webpage

DataBase Assignment



Figure 1: Snapshot of my webpage

B- Create a dBase. Make sure your database's name is your `YourFirstNameLastName`. The username for the database must be `db_YourFirstNameLastName`. For example database: `FaridFarahmand`
User: `db_FaridFarahmand`

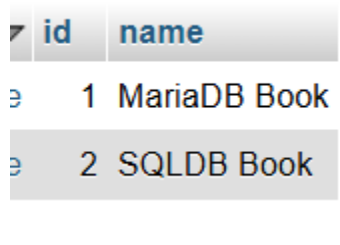
You should see something like this:

Database
information_schema
u301268454_AnthonyMeyer

Follow the Tutorial and create a simple `BOOK` table. Using `SHOW DATABASES` you can show your dBase. For more information Please refer to Github Tutorial. Your final `BOOK TABLE` should look something like this:

id	name
1	MariaDB Book
2	SQLDB Book

Take a snapshot of your results:



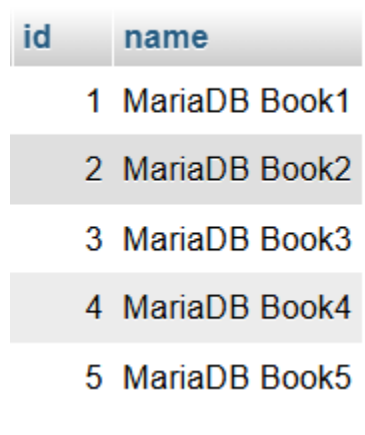
id	name
1	MariaDB Book
2	SQLDB Book

Figure 2: Snapshot of BOOK table.

1.2. Practice with MySQL Commands

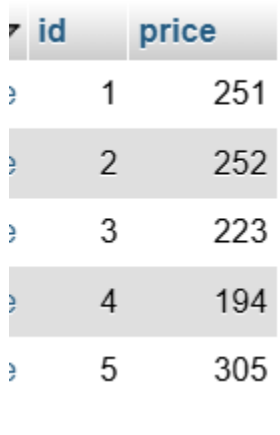
Using the phpMyAdmin in hostinger.com let's practice more with MySQL commands.

Review the commands in the database tutorial available on the course GitHub. Make sure you complete the tutorial. Once you complete the tutorial you should have a database table with your name and TWO tables having: `Price and Book`. Each table must have the appropriate elements.



id	name
1	MariaDB Book1
2	MariaDB Book2
3	MariaDB Book3
4	MariaDB Book4
5	MariaDB Book5

Figure 3a: Screenshot of Book table.



id	price
1	251
2	252
3	223
4	194
5	305

Figure 3b: Screenshot of Price table.

Now let's combine different fields and create ONE single table showing both book name and price - similar to below:

name	price
MariaDB Book1a	250
MariaDB Book2	252
MariaDB Book3	223
MariaDB Book4	194
MariaDB Book5	305

Take a snapshot of your results:

name	price
MariaDB Book1	251
MariaDB Book2	252
MariaDB Book3	223
MariaDB Book4	194
MariaDB Book5	305

Figure 4: Screenshot of joined Book and Price table

1.3. Create Your Own Sensory-Table

Now you need to design your own database using the following **design requirements**:

Design Two tables: `sensor_data & sensor_resister.`

`sensor_data` must have the following fields:

- node_name <up to 10 characters>; this will be used as the KEY
- time_received <indicating time/date>
- temperature <a numeric value between -10 to +100 celsius with maximum of 6 positions>
- humidity <a number value between 0-100 with maximum of 6 positions>

`sensor_register` must have the following fields:

- node_name <up to 10 characters>; this will be used as the KEY
- manufacturer <up to 10 characters>
- longitude < <a real (positive/negative number with maximum of 15 positions>
- Latitude <a real (positive/negative number with maximum of 15 positions>

You can find the longitude and latitude format here.

You can use a command similar to the following to create sensor data:

```
INSERT INTO sensor (node_name, time_received, temperature, humidity)
VALUES ('node-1','10-05-2022 10:05:34',84,25);
```

Your database must have the following requirements:

1- The starting data must be October 1, 2022, at 11:00 am.
2- The data must be sent every half an hour.
3- We need to register 5 different sensors with five different manufacturer names.
5- Each sensor must transmit at least 4 different data sets. This means there must be at least 4 different entries from each sensor node.
6- Node name format must be the following: node_1, node_2, node_3…. Etc.
7- No data from the sensor_data table is accepted if the transmitting node is not registered in the sensor_register table.
8- Create a VIEW of both tables and call it ensor_combined.

Once you have completed the database construction and the VIEW, take a snapshot of the content of the VIEW (sensor_combined) and all the commands you used to construct the databases. In addition, take a snapshot of the database structure, as it appears on the PhpMyAdmin portal, as shown below. Your snapshots should be similar to the figures below.

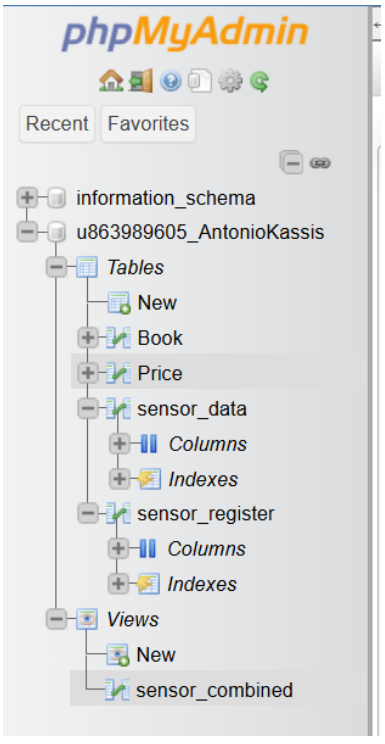


Figure 5a: Screenshot of the database structure

```sql
CREATE TABLE sensor_register (
    node_name VARCHAR(10) NOT NULL,
    manufacturer VARCHAR(10) NOT NULL,
    longitude DECIMAL(15,10) NOT NULL,
    latitude DECIMAL(15,10) NOT NULL,

    PRIMARY KEY (node_name)
);

CREATE TABLE sensor_data (
    id INT NOT NULL AUTO_INCREMENT,
    node_name VARCHAR(10) NOT NULL,
    time_received DATETIME NOT NULL,
    temperature DECIMAL(6,2) NOT NULL,
    humidity DECIMAL(6,2) NOT NULL,

    PRIMARY KEY (id),

    UNIQUE (node_name, time_received),

    CONSTRAINT fk_sensor_node
        FOREIGN KEY (node_name)
        REFERENCES sensor_register(node_name),

    CHECK (temperature >= -10 AND temperature <= 100),
    CHECK (humidity >= 0 AND humidity <= 100)
);
```

Figure 5b: Screenshot of the table commands

node_name	manufacturer	longitude	latitude	time_received	temperature	humidity
node_1	Bosch	-122.714100	38.339600	2022-10-01 11:00:00	22.500	45.000
node_1	Bosch	-122.714100	38.339600	2022-10-01 11:30:00	23.000	46.000
node_1	Bosch	-122.714100	38.339600	2022-10-01 12:00:00	23.500	47.000
node_1	Bosch	-122.714100	38.339600	2022-10-01 12:30:00	24.000	48.000
node_2	Siemens	-122.713500	38.340100	2022-10-01 11:00:00	20.000	50.000
node_2	Siemens	-122.713500	38.340100	2022-10-01 11:30:00	20.500	51.000
node_2	Siemens	-122.713500	38.340100	2022-10-01 12:00:00	21.000	52.000
node_2	Siemens	-122.713500	38.340100	2022-10-01 12:30:00	21.500	53.000
node_3	Honeywell	-122.712900	38.340600	2022-10-01 11:00:00	25.000	40.000
node_3	Honeywell	-122.712900	38.340600	2022-10-01 11:30:00	25.500	41.000
node_3	Honeywell	-122.712900	38.340600	2022-10-01 12:00:00	26.000	42.000
node_3	Honeywell	-122.712900	38.340600	2022-10-01 12:30:00	26.500	43.000
node_4	DHT	-122.712300	38.341100	2022-10-01 11:00:00	27.000	38.000
node_4	DHT	-122.712300	38.341100	2022-10-01 11:30:00	27.500	39.000
node_4	DHT	-122.712300	38.341100	2022-10-01 12:00:00	28.000	40.000
node_4	DHT	-122.712300	38.341100	2022-10-01 12:30:00	28.500	41.000
node_5	Sensirion	-122.711700	38.341600	2022-10-01 11:00:00	18.000	55.000
node_5	Sensirion	-122.711700	38.341600	2022-10-01 11:30:00	18.500	56.000
node_5	Sensirion	-122.711700	38.341600	2022-10-01 12:00:00	19.000	57.000
node_5	Sensirion	-122.711700	38.341600	2022-10-01 12:30:00	19.500	58.000

Figure 5c: Screenshot of the sensor_combined VIEW's values

Prove that if you insert data for a sensor that is NOT registered the value will not be accepted.

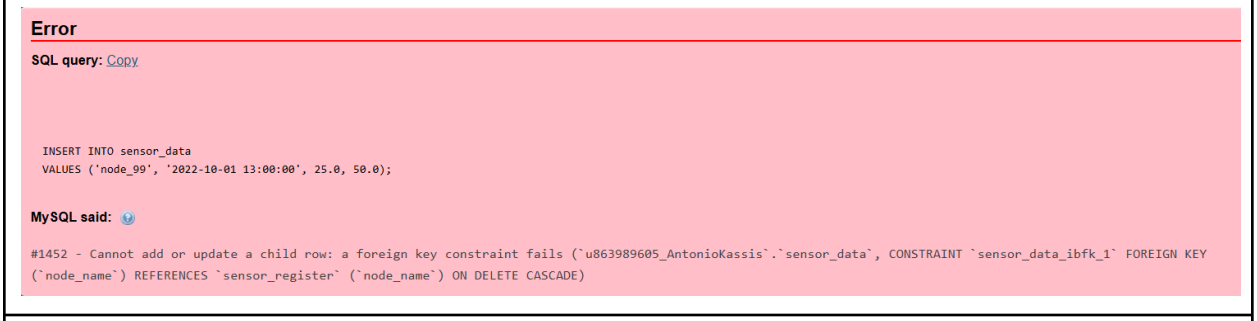


Figure 6: Screenshot of the constraints stopping an invalid tuple into the table

Please note: During your demonstration, you must show that all assigned features are fully functional. Each missing or non-functioning feature will result in a **7-point deduction** from your grade.

Then, in the provided space below, provide the link to that web page. This page will serve as your **main web page for the remainder of the semester**, and all assignment links must be posted there. This is a simple template I have created: https://faridfarahmand.net/SampleStudentAssignmentIndex.html

You can copy the template page and easily update it.

https://antoniokassis.com/assignments.html
Your web page link.

2- Answer the following questions

Answer the following questions.

1. What type of database does hostinger.com use?
   Hostinger uses MariaDB, which is a relational database management system that is compatible with MySQL.

2. What type of web server does hostinger.com use?
   Hostinger uses the LiteSpeed Web Server.

3. Referring to the data type tutorial (https://www.tutorialspoint.com/mariadb/mariadb_data_types.htm) assuming we have the following:
   ```
   CREATE TABLE ints (a INT,b INT UNSIGNED,c INT ZEROFILL);
   ```

   Would this insertion command work? `INSERT INTO ints VALUES (-10,-10,-10);`
   No. The value -10 can be stored in a, but b is UNSIGNED, so it cannot store negative values. ZEROFILL is also intended for non-negative values.

4. What is the purpose of having a primary key in a dBase?
   A primary key uniquely identifies each row in a table and prevents duplicate records from using the same key value.

5. What is the purpose of having a foreign key in a dBase?
   A foreign key creates a relationship between two tables and helps make sure referenced data exists in the related table.

6. Can a database have multiple tables?
   Yes. A database can contain many tables, and those tables can be related to each other.

7. What is a relational database?
   A relational database stores data in tables made of rows and columns. The tables can be linked together using keys.

8. Is MariadBase a relational or non-relational dBase?
   MariaDB is a relational database.

9. What is a database schema?
   A database schema is the structure or design of a database. It defines the tables, columns, data types, keys, and relationships.

10. What is a Table in a relational Database?
    A table is a collection of related data organized into rows and columns.

11. What are the main components of a Table in a relational database?
    The main components are rows, columns, data types, keys, and constraints.

12. In terms of data type, what is the difference between char(size) and varchar(size)?
    CHAR(size) stores fixed-length text, while VARCHAR(size) stores variable-length text up to the maximum size specified.

13. What would be an example of datetime datatype format?
    YYYY-MM-DD HH:MM:SS

3. GitHub

- Please post your code (PHP, SQL, etc.) in your GitHub. Make sure it is clear which assignment the code belongs to. All codes must have comments.
- Please add the link to your GitHub repository on the web page you created in **Section 1.1**.

Link to your GitHub: https://github.com/Neo4724/IoT_EE_470/tree/main/EE%20470%20-%20Assignment%204%3A%20Part%201%20-%20[PHP%20%26%20Database]

4. Quick Survey

Answer the following questions, please:

Survey question	Response
On a scale of 1-10 how did you like this exercise? (10 is the best, 1 is the worst)	8
On a scale of 1-10 how much did you learn as a result of completing this exercise? (10 = plenty; 1=very little)	7
How many hours did you spend completing this exercise?	9

END OF PART 1